

Applied Spectral Graph Theory: Single-Cell Manifold Learning & Network Connectivity

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Introduction

Single-cell RNA sequencing measures thousands of genes across thousands of cells. The challenge is that biology is not arranged in neat rows and columns: it lives on a high-dimensional shape, or manifold, where nearby cells often share identity, state, or trajectory.

My independent study asked how spectral graph theory can recover that hidden shape. I treated cells as nodes, connected similar cells with weighted edges, and used the graph Laplacian to transform raw expression into a map of structure.

Big idea: graph eigenvectors can turn a cloud of measurements into interpretable communities.

Add Image 1

manual graph or Laplacian

Add Image 2

NetworkX or heat map

The Core Mathematics

To study a graph with linear algebra, I built three matrices:

- Adjacency matrix A: stores which nodes are connected, and by how strongly.
- Degree matrix D: records how much total edge weight touches each node.
- Laplacian matrix $L = D - A$: acts like a discrete curvature or diffusion operator.

The eigenproblem $L v = \lambda v$ reveals the graph's natural modes. Small eigenvalues describe broad, smooth structure; their eigenvectors provide coordinates for clustering without assuming the groups in advance.

For biological data, normalized Laplacian ideas help keep high-degree cells from dominating the geometry unfairly.

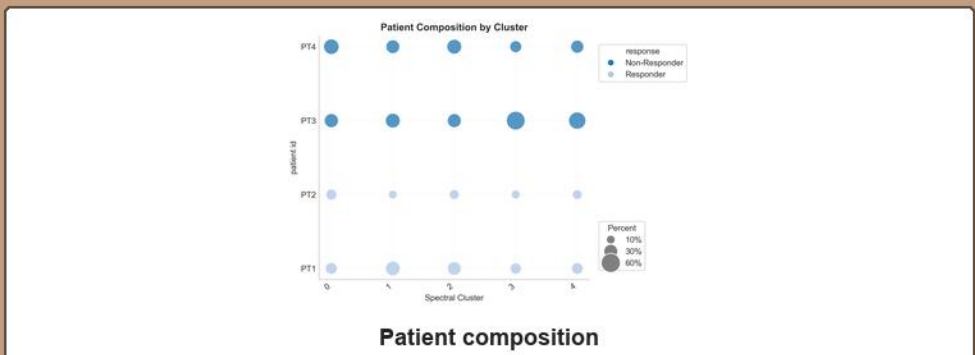
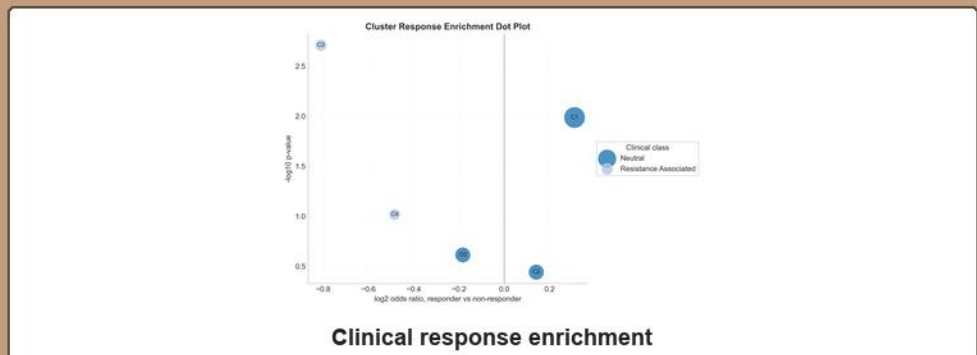
Why It Matters

Spectral methods connect pure linear algebra, network science, and biological discovery. A single eigenvector can identify a bottleneck in a social network, a weak point in infrastructure, or a boundary between cell states.

In medicine, this matters because hidden immune-cell communities may help explain why some patients respond to treatment and others do not.

Applications

- Finding communities in large networks without labels.
- Measuring robustness through algebraic connectivity.
- Reducing high-dimensional biological data while preserving local neighborhoods.
- Linking computational clusters to clinical response hypotheses.



The Fiedler Value & Robustness

The smallest eigenvalue of a connected graph Laplacian is always $\lambda_1 = 0$. The second smallest eigenvalue, λ_2 , is the Fiedler value, also called algebraic connectivity.

If $\lambda_2 > 0$, the graph is connected. If it is close to zero, the graph has a narrow bridge or bottleneck. Cheeger's inequality connects this eigenvalue to the graph's minimum cut, so connectivity becomes measurable rather than just visual.

Add Image 3

targeted attack curve

Add Image 4

spectral gap diagnostic

Case Study: Spectral Clustering

I applied the pipeline to a breast-cancer immunotherapy case study. Each cell was embedded in gene-expression space, connected to its nearest transcriptional neighbors, and clustered using eigenvectors of the graph Laplacian.

2,500

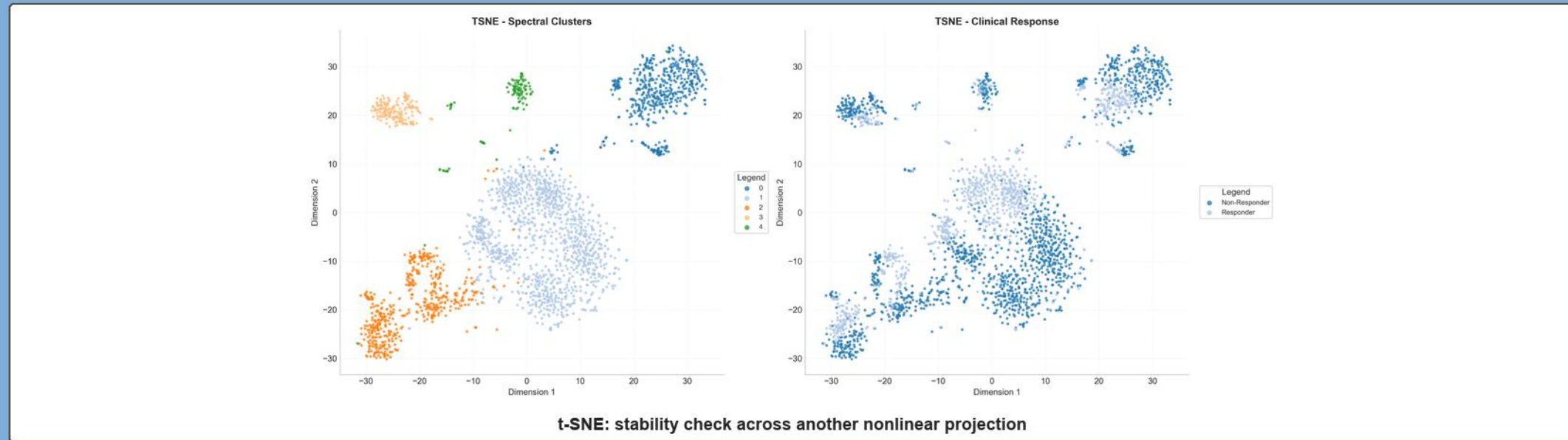
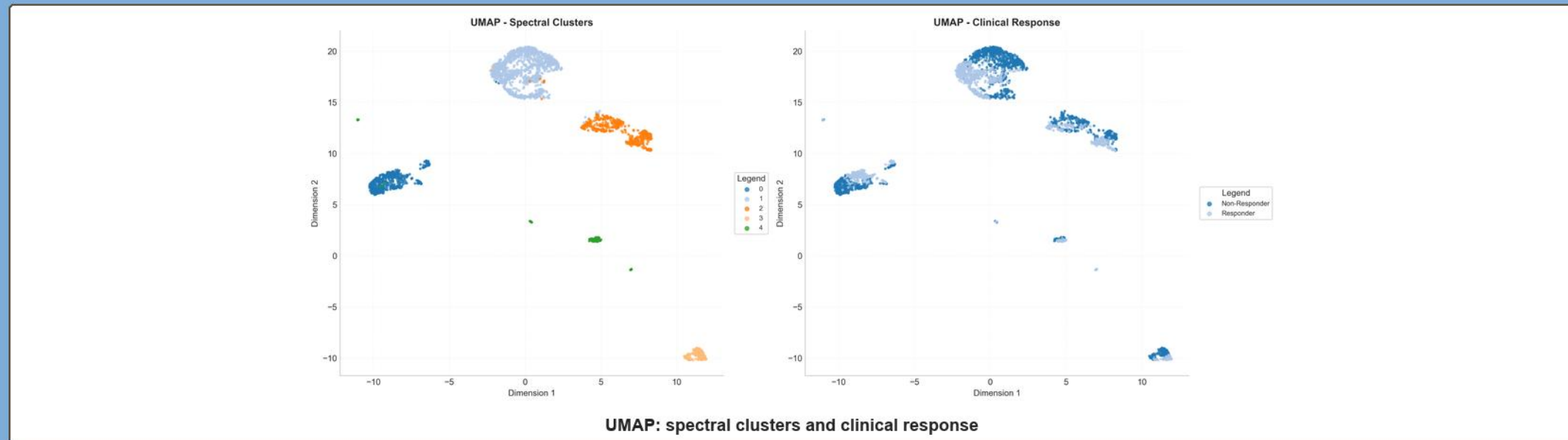
cells analyzed

5

spectral communities

2

response views



Results & Interpretation

The spectral clustering pipeline isolated distinct cell communities across patient cohorts. The clusters were not assigned by hand; they emerged from graph connectivity.

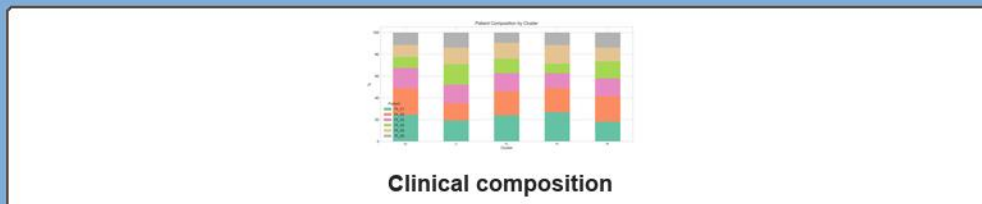
In the Week 9 readout, one T-cell-dominant community appeared response-associated, while a myeloid-dominant community appeared resistance-associated. That is the exciting bridge: an eigenvector calculation became a biological hypothesis about immune response.

- Response signal: T-cell cluster, fold ratio 1.45, $p = 7.24e-05$.
- Resistance signal: myeloid cluster, fold ratio 0.60, $p = 1.39e-04$.

More Figures to Add

This panel is the easiest place to swap in supporting visuals without disturbing the main story.

- Full clinical cluster importance bar chart.
- Cell-type composition or patient share chart.



The Research Journey

This project began with small graphs and hand calculations. I manually built adjacency, degree, and Laplacian matrices so the code would not feel like a black box.

The hardest shift was conceptual: moving from geometric intuition in ordinary space to structure in thousands of dimensions. The breakthrough came when the abstract math produced visible communities in the biological data.

Weeks 1-2

Matrix foundations and manual Laplacian derivations.

Weeks 3-4

Fiedler values, centrality, heat maps, and robustness.

Weeks 6-8

Stress tests, heat kernels, and higher-order topology ideas.

Week 9

Single-cell case study and poster-ready analysis.

Next Steps

Future work could test the pipeline on more real datasets, compare spectral clustering with graph neural networks, and extend from graph Laplacians to Hodge Laplacians for modeling multi-way cell interactions.

Acknowledgement: thank you to my Independent Study advisor/content expert for helping shape the scope, math focus, and presentation feedback.